

Neovascular glaucoma after proton beam therapy of choroidal melanoma: incidence and risk factors

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Abstract

Purpose To analyze the risk factors for the development of neovascular glaucoma (NVG) of patients with choroidal melanoma after proton beam therapy (PBT).

Method Clinical case series, retrospective study. We evaluated 629 consecutive patients receiving proton beam therapy for the treatment of a choroidal melanoma at the oncology service at Charité, Berlin and Helmholtz-Zentrum, Berlin between 05/1998 and 11/2008 regarding the development and risk factors of NVG. Patients with tumor resection, salvage proton beam therapy for recurrent disease and known glaucoma of other origin were excluded from the cohort.

Results Of the 629 patients matching the inclusion criteria, 20.8% developed neovascularization of the iris after a mean time of 2.0 years (range 0.45 to 8.4 years) after PBT. Forty-seven percent of the patients with a neovascularization of the iris developed NVG after a mean time of 2.0 years after PBT, ranging from 5 months to 11.6 years. Univariate analysis revealed tumor height [$p < 0.001$, hazard ratio (HR): 2.71, 95% confidence interval (CI): 1.36–5.35 for tumors >6 mm ≤ 9 mm and 11.32 (4.03–31.73) for tumors >9 mm], distance of the tumor to the optic disc ($p < 0.001$, HR: 0.43, 95% CI: 0.24–0.77 for >0 mm ≤ 3 mm and HR: 0.13, 95% CI: 0.04–0.37 for >3 mm), dose to the ciliary body ($p < 0.001$, HR: 9.21, 95%

CI: 5.08–16.71 (21–40 cobalt gray equivalents (CGE), HR 27.23, 95% CI: 6.33–116.97 (41–60 CGE)), dose to the optic disc ($p < 0.001$, HR: 3.53, 95% CI: 1.11–11.27 (21–40CGE), HR: 5.37, 95% CI: 2.72–10.63 (41–60CGE)), the irradiated length of the optic nerve ($p < 0.001$, HR: 4.48, 95% CI: 2.47–8.13) and diabetes mellitus ($p < 0.05$, HR: 2.53, 95% CI: 1.4–4.5) were found to be risk factors for the development of NVG. Multivariate regression analysis identified the dose to the ciliary body [$p < 0.001$, HR: 4.39, 95% CI: 2.28–8.44 (21–40 CGE), HR: 11.04, 95% CI: 1.97–61.69 (41–60 CGE)], the irradiated length of the optic nerve ($p < 0.001$, HR: 3.88, 95% CI: 2.11–7.16), the existence of diabetes mellitus ($p < 0.01$, HR: 1.28, 95% CI: 1.24–4.21) and tumor height [$p < 0.05$, HR: 2.28, 95% CI: 1.17–4.83 (>6 mm ≤ 9 mm), HR: 3.74, 95% CI: 1.05–13.23, (>9 mm)] to be independent risk factors for the development of NVG.

Conclusions In the present analysis we found tumor height, dose to the ciliary body, irradiated length of the optic nerve and diabetes mellitus to be risk factors for the development of NVG. Whenever possible, critical structures of the anterior and posterior segment should be spared by beam shaping or changing of the beam entry angle.

Keywords Neovascular glaucoma · Irradiation-induced glaucoma · Proton beam therapy · Choroidal melanoma

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Introduction

Uveal melanoma, although the most common primary malignant melanoma in the eye in adults, is a rare disease [1]. During recent decades treatment options have significantly improved. By means of radiation, high local tumor control rates can be achieved [2–4]. Adjuvant therapies such as endoresection or transscleral resection allow the treatment of

large tumors which otherwise would have been enucleated. In addition to local tumor recurrence, NVG is the most common reason for secondary enucleation in tumor-bearing eyes [5, 6]. NVG can be found after brachytherapy (e.g. Ru-106, I-125) [2, 6] and teletherapy (e.g. proton beam therapy, carbon ion therapy) [7–9]. Depending on the tumor configuration and the method of radiation, NVG can be found in 7.0% [10] up to 42.8% [9]. Different studies have shown that there are tumor associated risk factors, e.g. tumor size and location, as well as radiation associated risk factors, e.g. dose to the ciliary body for the development of NVG [7, 9]. However, at the moment detailed pathological mechanisms of the development of NVG are not understood.

Ferry et al. were one of the first authors describing the histology of iris neovascularization after PBT. Today it is believed that the development of neovascularization is a result of an imbalance between pro-angiogenic and anti-angiogenic factors. Elevated levels of VEGF-A, a pro-angiogenic cytokine, have been found in the aqueous and vitreous body of uveal melanoma patients [11]. It has been shown that elevated VEGF levels correlate with previous radiation therapy and with the presence of neovascularization of the iris or optic disc.

NVG is a known issue after ischaemic alterations and can also be found after retinal vein occlusion or as a result of diabetic retinopathy. In these diseases the use of laser coagulation of the ischaemic areas, as well as the use of anti-VEGF and corticoid derivatives, led to a massive improvement in morbidity. These methods are increasingly used for the treatment of radiation retinopathy and for prophylaxis as well as causative based treatment of NVG [4].

In this study we analyzed the predictive risk factors for the development of NVG after PBT.

Materials and methods

In this retrospective study we included all patients with choroidal melanoma treated with proton beam therapy at the Helmholtz-Zentrum, Berlin using a 68 megaelectron volts (MeV) fixed proton beam between 05/1998 and 11/2008 [12, 13].

The patients received surgical tantalum clip placement before proton beam radiation.

All patients received a total dose of 60 CGE (cobalt gray equivalents), which were delivered in daily fractions of 15 CGE on four consecutive days. The clinical target received 100% of the dose. A safety margin of 1.00 mm (90% isodose) was added. Treatment planning was made with EYEPLAN software version 1.2 [14] and since 2006 with OCTOPUS software version 4.4. Information regarding technique and dosimetry has been published before [15].

We excluded all patients with adjuvant endoresection or transscleral resection, glaucoma before proton beam therapy and those patients receiving proton beam therapy as salvage therapy for local recurrence. In total we included 629 patients in our study. Patients were examined at our clinic depending on the clinical course of the disease.

Based on the definition of the European glaucoma society, NVG was defined as persisting intraocular pressure of more than 21 mmHg in patients with an iridotrabecular fibrovascular membrane without known history of glaucoma of another origin [16]. Radiation retinopathy, optic neuropathy and local recurrence were defined as previously described [3].

The tumors were categorized according to the TNM classification for choroidal melanoma [17]. The study was approved by the Institutional Review Board of the Charité-Universitätsmedizin Berlin, Germany, and was in accordance with the Declaration of Helsinki.

Statistical analysis

Data were entered into a database and cleaned and checked for errors. Dose concentrations were split into categories of 20 CGE. Follow-up periods were aligned using the date of initial radiation treatment as the starting date. Kaplan-Meier analysis was used to estimate overall survival and ocular outcome. We used a univariate Cox-regression to identify possible risk factors. Factors with strong evidence for an effect were entered successively into a multivariate Cox-regression starting with variables that showed the strongest evidence for an effect. Risk factors that showed no evidence for an effect in the multivariate analysis were excluded from the model. A *p*-value of <0.05 was considered strong evidence for an effect. Statistical analysis was done using STATA version 12.1 (StataCorp, College Station, TX, USA) and graphs were drawn using the ggplot2 module in R-statistics.

Results

Baseline characteristics

We identified 629 patients with choroidal melanoma treated with proton beam therapy. The mean follow-up time was 5.1 years (range: 0.3–14.2 years). The mean age was 59.3 years (standard deviation [SD] 13.2 years, range 16–88 years) at the time of proton beam therapy. Of these patients, 52.2% were male, 47.9 were female, and 13.7% were diabetic. For basic tumor characteristics see Table 1. According to the TNM classification 38.3% of the patients were categorized as T1, 47.7% as T2, 12.7% as T3, and 1.3% as T4. In Kaplan-Meier analysis the 5-year overall survival rate was 94% (92–96%), metastasis-free survival 90% (87–92%), and tumor-associated death was 3% (1.8–4.9%).

Table 1 Tumor and irradiation characteristics of choroidal melanoma treated with proton beam therapy

Variable	mean	SD	minimum	maximum
Tumor height [mm]	3.8	1.7	1.3	14.1
Largest tumor diameter [mm]	11.0	2.9	3	20.5
Distance to fovea [mm]	1.1	1.6	0	12.4
Distance to optic disc [mm]	1.8	2.1	0	13.4
Dose to fovea [CGE]	39.9	25.5	0	60
Dose to optic disc [CGE]	29.1	27.4	0	60
Irradiated length of the optic nerve [mm]	1.2	1.4	0	6

SD standard deviation, CGE cobalt gray equivalents

Neovascularization of the iris and NVG

Of the 629 patients matching the inclusion criteria 131 (=20.8%) developed neovascularization of the iris after a mean time of 2.0 years (range 0.45 to 8.4 years). Sixty-two (=47.3%) of the patients with neovascularization of the iris developed NVG. The mean time until the development of NVG was 2.0 years, ranging from 5 months to 11.6 years. The five-year Kaplan-Meier estimation for NVG was 10.5% (95% CI: 8–13.5%) (Fig. 1). Enucleation was carried out in 25 patients (=4.0%). Twelve patients (=1.91%) were enucleated because of NVG, ten patients (=1.25%) because of local recurrence, two patients (=0.32%) because of phthisis bulbi and one patient (=0.16%) after multiple vitreoretinal surgeries, amaurosis, and decompensation of intraocular pressure without NVG.

Of the patients with NVG 68.3% received laser coagulation from once to up to eight times after a median of 17 months $\pm$ 5.7. Cyclocryocoagulation and cyclophotocoagulation (median: 1, SD 0.7, range: 1–3) were performed in 51.7% after a median time of 38 months $\pm$ 5.0.

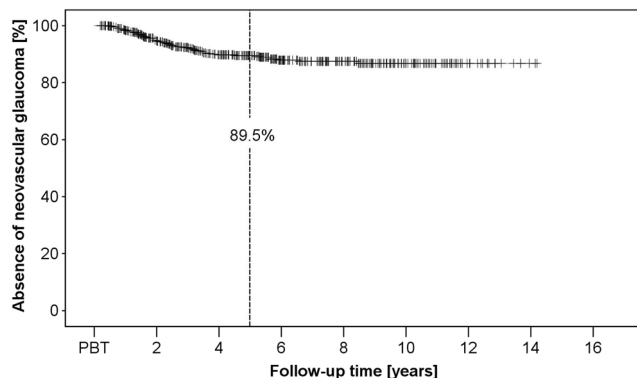


Fig. 1 Kaplan-Meier analysis for the absence of neovascular glaucoma for all patients treated with proton beam therapy for choroidal melanoma. PBT: Proton beam therapy

Radiation-induced retinopathy and optic neuropathy

For all patients the 5-year Kaplan-Meier estimator was 14.2% for the absence of radiation-induced retinopathy and 36.6% for the absence of optic neuropathy. The 5-year Kaplan-Meier estimator for the absence of radiation-induced retinopathy was 3% for patients with NVG and 15.5% for patients without NVG. The 3-year Kaplan-Meier estimator for the absence of optic neuropathy was 6.1% for patients with NVG and 32.1% for patients without NVG. The log-rank test showed $p < 0.001$ comparing the patients with NVG and without NVG regarding the absence of radiation-induced retinopathy and optic neuropathy.

Risk factors for the development of NVG

Univariate analysis revealed tumor height ($p < 0.001$) and the distance of the tumor to the optic disc ($p < 0.001$) as tumor-based risk factors. In contrast, the largest tumor diameter, retinal detachment and the distance of the tumor to the fovea were not risk factors. Irradiation parameters with a significant risk for the development of NVG were the dose to the ciliary body, the lens, the optic disc, and the irradiated length of the optic nerve (all: $p < 0.001$). Diabetes mellitus was found to be a tumor and irradiation independent risk factor for the development of NVG ($p < 0.05$).

In the next step the variables which were considered statistically significant in univariate analysis were entered into a multivariate regression analysis. Dose to the ciliary body and irradiated length of the optic nerve were found to be independent risk factors for the development of NVG to a highly significant level ($p < 0.001$). Furthermore, the existence of diabetes mellitus ($p < 0.01$) and tumor height ($p < 0.05$) were significant risk factors (Table 2).

Discussion

In the present study the 5-year cumulative NVG rate was 10.5% after the treatment of choroidal melanoma with proton beam therapy. This is comparable to the results found by other groups for the treatment of uveal melanoma by the means of PBT (12.7%/5y [7]) or plaque radiotherapy (15.0%/5y [6]). However, it is significantly less compared to the risk of NVG after carbon ion radiotherapy (3-year cumulative NVG rate: 42.6%, [9]). Siedlecki et al. compared the incidence of secondary glaucoma after treatment of uveal melanoma with CyberKnife robotic radiosurgery versus Ruthenium-106 brachytherapy [18]. They report an incidence of secondary glaucoma of 46.7% 5 years after CyberKnife robotic radiosurgery and 11.1% after Ruthenium-106 brachytherapy. The incidence of rubeosis iridis was 30.6% and 5.3%, respectively. Unfortunately, the incidence of NVG has not been explicitly

Table 2 Univariate and multivariate analysis of potential risk factors for the development of neovascular glaucoma after proton beam therapy of choroidal melanoma

	n/N	Hazard ratio (95% CI)	p-value	Adjusted hazard ratio	p-value
Age	629/629	1.01 (0.98–1.03)	0.372		
Sex					
Male	29/328	1			
Female	33/301	1.22 (0.75–2.02)	0.418		
Diabetes mellitus					
No	48/543	1		1	
Yes	14/86	2.53 (1.4–4.5)	0.0015	1.28 (1.24–4.21)	0.008
Retinal detachment					
No	22/250	1			
Yes	40/379	1.31 (0.78–2.21)	0.308		
Tumor height					
≤ 6 mm	48/566	1		1	
> 6 mm ≤ 9 mm	10/55	2.71 (1.36–5.35)		2.28 (1.17–4.83)	
> 9 mm	4/8	11.32 (4.03–31.73)	<0.001	3.74 (1.05–13.23)	0.023
Largest tumor diameter					
≤ 10 mm	20/250	1			
> 10 mm ≤ 16 mm	37/352	1.44 (0.82–2.48)			
> 16 mm	5/27	3.19 (1.19–8.52)	0.084		
Distance to fovea					
0 mm	27/333	1			
> 0 mm ≤ 3 mm	28/122	1.66 (0.98–2.82)			
> 3 mm	7/74	1.16 (0.51–2.66)	0.166		
Distance to optic disc					
0 mm	42/247	1			
> 0 mm ≤ 3 mm	16/217	0.43 (0.24–0.77)			
> 3 mm	4/165	0.13 (0.04–0.37)	<0.001		
Dose to the ciliary body					
0–20 CGE	45/589	1		1	
21–40 CGE	15/37	9.21 (5.08–16.71)		4.39 (2.28–8.44)	
41–60 CGE	2/3	27.23 (6.33–116.97)	<0.001	11.04 (1.97–61.69)	<0.001
Dose to lens*					
0–20 CGE	39/557	1			
21–40 CGE	13/36	8.37 (4.43–15.82)			
41–60 CGE	6/14	9.98 (4.21–23.74)	<0.001		
Dose to fovea					
0–20 CGE	18/188				
21–40 CGE	7/41	1			
41–60 CGE	37/400	1.68 (0.71–4.04)	0.446		
Dose to optic disc					
0–20 CGE	10/306	1			
21–40 CGE	4/39	3.53 (1.11–11.27)			
41–60 CE	48/284	5.37 (2.72–10.63)	<0.001		
Irradiated length of the optic nerve					
0 mm	14/353			1	
> 0 mm	48/276	4.48 (2.47–8.13)	<0.001	3.88 (2.11–7.16)	<0.001

CI confidence interval, CGE cobalt gray equivalents

*Missing values in 22 observations

described. Above all, one has to consider the fact that Ruthenium-106 brachytherapy is usually limited to tumors with a maximum height of less than 6.0 mm, when comparing the different treatment modalities.

Mishra et al. found a 5-year NVG rate of 12.7% after PBT of uveal melanoma [7], which is only slightly more than in our study. This could be due to our population showing a smaller mean tumor height in comparison to Mishra (5.2+/-2.7 mm vs. 3.8 +/-1.7 mm). A reason for this difference could be that Mishra et al. only excluded patients with tumor resection before PBT. We excluded patients with adjuvant and neoadjuvant tumor resection in our study as it could be shown by different groups that tumor resection significantly reduces the risk for NVG [19]. Cassoux et al. were able to show that the risk for NVG and secondary enucleation was significantly reduced in patients receiving PBT and endoresection in comparison to those patients receiving PBT alone or PBT in combination with TTT (transpupillary thermotherapy) [19].

Patients receiving tumor resection usually have larger tumors. Tumor height was the only tumor-associated risk factor for the development of NVG in multivariate analysis in our study. Foss et al. had already shown in 1997 that large tumor size is a significant risk factor for the development of rubeosis [8]. Mishra et al. found tumor height to be the most important predictor of NVG in multivariate analysis after PBT of uveal melanoma [7]. In contrast to our results the largest tumor diameter was also a risk factor in their analysis, even though both studies had comparable tumor diameters (11.0 +/-2.9 mm in our study vs. 10.7 +/- 3.4 mm Mishra et al.) [7]. Similar to our findings, tumor location was not a significant risk factor in their analysis. Sagoo et al. also found mean tumor thickness to have the highest predictive value for radiation-induced NVG after the treatment of juxtapapillary choroidal melanoma by means of plaque radiotherapy (Ruthenium-106, Iodine-125, Cobalt-60, or Iridium-192) [6]. Siedlecki et al. found tumor thickness to be strongly associated with the incidence of secondary glaucoma ($p < 0.001$), but not with tumor localization or ciliary body involvement in their analysis of uveal melanoma treated with CyberKnife robotic radiosurgery or Ruthenium-106 brachytherapy [18].

Radiation-dependent risk factors for the development of NVG were the irradiated length of the optic nerve ($p < 0.001$) and the dose to the ciliary body ($p < 0.001$). Mishra et al. also found the irradiated length of the optic nerve to be a risk factor for NVG, even though they were using other cut-off values in their study (≤ 1 mm vs. > 1 mm, $p = 0.02$). In comparison to our study they also found that the disc treated to $\geq 50\%$ dose is an independent risk factor. This was only significant in univariate analysis in our study, but not in multivariate analysis. Radiation of these structures might lead to occlusion of the central retinal vein, artery or their major branches, leading to posterior ischemia [7]. The dose to the optic disc, as well as the optic nerve is mainly determined by

the tumor location. Therefore, sparing of these structures is usually only possible to a limited extent.

The dose to structures of the anterior segment has been found to be a risk factor for the development of NVG after PBT by several groups with different irradiation techniques, e.g. carbon ion radiotherapy [9]. In comparison to our results Mishra et al. did not find the dose to the ciliary body to be a risk factor for the development of NVG after PBT, but the dose to the lens was [7]. The lens is a metabolically inactive structure and it is unlikely to be directly involved in the development of NVG itself. It is more probable that the dose to the lens is highly correlated to the dose to the anterior segment, especially the dose to the iris. Usually the dose to the iris is not calculated during the planning procedure because of its high correlation to the dose to the lens, as well as to its variable size. We recently published our experience with trabeculectomy after proton beam therapy of uveal melanoma. In this cohort, 14 out of 15 tumors were of the anterior segment involving either the iris and/or the ciliary body. Seven of these patients developed neovascularization of the iris even though there was no significant dose to posterior structures [20]. Furthermore, the appearance of neovascularization is not necessarily limited to the irradiated area, but can also appear in the untreated part of the iris as already mentioned in the introduction [21]. It is probable that neovascularization of the iris might also develop from radiation induced ischaemia of the anterior segment or local inflammation and not only as a consequence of retinal ischaemia. We, therefore, recommend avoiding the irradiation of the anterior segment by beam shaping or changing of the beam entry angle if possible.

Diabetes mellitus was found to be a radiation and tumor independent risk factor for the development of NVG in our study ($p < 0.05$). Histologic alterations and pathologic mechanisms of diabetic and radiation retinopathy are quite similar and synergism between diabetic and radiation retinopathy has been discussed for a while [22]. Even subclinical alterations as a consequence of a diabetes mellitus could trigger the development of radiation retinopathy. In this present study there was no significant difference between the time of diagnosis of NVG regarding the absence or presence of diabetes mellitus (mean time to NVG in patients with diabetes mellitus: 23.0 months, without diabetes mellitus: 24.6 months). However, one of the limitations of our study is that we did not perform a grading of diabetic retinopathy before radiation, or perform a grading of radiation retinopathy.

In most studies the mean time until the diagnosis of NVG is about 2 to 3 years [7, 9]. In our study, the time until NVG was 2.0 years, range: 5 months to 11.6 years. Apart from that we found a very short time frame between the development of iris neovascularization and NVG (time until neovascularization: 2.0 years, range: 0.45 to 8.4 years). The fact that there seems to be only a short time interval between the development of iris neovascularization and the development of a NVG shows

the importance of regular checks of the intraocular pressure and for signs of ischaemia, especially during the first 3 years after irradiation and an early and effective treatment of ischaemia to avoid the development of NVG.

It is interesting that there is still a broad range for the time of development of NVG, with some patients developing a kind of an early onset NVG after very few months and some patients developing a late onset NVG many years after PBT. Mishra et al. found the mean time to diagnosis of NVG to be 22 months after PBT of uveal melanoma, showing a broad range from 3.6 to 87 months, which is comparable to our results [7]. At the moment it is not known what triggers the time of development for NVG in different patients. As already mentioned above, diabetes mellitus was not a factor influencing the time until NVG diagnosis in this study. Genetic factors might play a role, e.g. it is known that patients with ataxia-telangiectasia are highly sensitive to radiation and can develop severe radiotoxic side effects. The ATM (ataxia-telangiectasia mutated) gene is homozygously mutated in patients suffering from ataxia-telangiectasia. Heterozygous mutations of ATM appear in up to 1% in population [23]. There is evidence that a heterozygous ATM status has an intermediate status between ataxia-telangiectasia patients and general population regarding radiosensitivity, as well as cancer predisposition [23]. Ostendorf et al. report about an allogeneic transplant recipient with a heterozygous ATM mutation developing severe radiotoxicity with dermatitis, colitis and radiation retinopathy 30 days post-transplantation after 6×2 Gray total body irradiation on days -7, -6, and -5 [24]. It has been shown that ATM heterozygous mice are more sensitive to radiation-induced cataracts in comparison to their wild-type counterparts [25]. However, at the moment the results are controversial and data, especially on radiation retinopathy, are still lacking.

In conclusion NVG is still a major issue after irradiation-based treatment of uveal melanoma. The development of NVG is most probably multifactorial. In this study, tumor size, dose to the ciliary body, irradiated length of the optic nerve and the presence of diabetes mellitus were found to be independent risk factors for the development of NVG. Whenever possible, critical structures of the anterior and posterior segment should be spared by beam shaping or changing of the beam entry angle.

Compliance with ethical standards

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Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional

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Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent For this type of study formal consent is not required.

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